December 2015 - F70LA

Solutions

Q1 (a) $q_{71} = 1 - p_{71}$

$= 1 - \frac{155-71}{100} = \underline{0.16}$

(b) ${}_3p_{70} = p_{70}\, p_{71}\, p_{72} = \underline{0.59262}$

(c) $q_{74} = 0.19$

$\therefore {}_1q_{75} = 0.81 \times 0.2 = 0.162$

$A^1_{74:2\overline{1}} = 0.19v + 0.162v^2 = 0.334804$

Value $= £2m \times 0.334804 = \underline{£669,607}$

(d) ${}^2A$ is calculated on $(2i + i^2) = 7.1225\%$

${}^2A^1_{74:2\overline{1}} = 0.19v + 0.162v^2$ (again, but on ↗) $= 0.318541$

Variance $= {}^2A - (A)^2 = 0.318541 - (0.334804)^2$

$= 0.206447$

SD $= \sqrt{\quad} = 0.454365$

$\times £2m$ gives $\underline{£908,729}$

Q2

(a) (i) $S_0(0) = e^0 = 1$ ✓

(ii) $\lim_{x\to\infty} S_0(x) = "e^{-\infty}" = 0$ ✓

(iii) $\frac{dS}{dx} = \left(-\frac{1}{20} - \frac{3x^2}{300{,}000}\right) S(x)$

< 0 ✓

(b) ${}_{40}p_0 = S_0(40) = \underline{0.10934}$

(c) ${}_tp_x = \frac{S_0(x+t)}{S_0(x)}$

$= \exp\left(\frac{x}{20} - \frac{(x+t)}{20} + \frac{x^3 - (x+t)^3}{300{,}000} \right)$

$= \exp\left(-\frac{t}{20} - \frac{3x^2t + 3xt^2 + t^3}{300{,}000} \right)$

Q3 (a) (i) $_2p_{[40]} = p_{[40]}\, p_{[40]+1}$

$= p_{36}\, p_{39}$

$= 0.99925 \times 0.99902 = \underline{0.99827}$

(ii) $_3p_{[40]+1} = p_{[40]+1}\, p_{42}\, p_{43}$

$= p_{39}\, p_{42}\, p_{43} = \underline{0.99631}$

(iii) $_3|q_{[40]}$ $= {}_3p_{[40]}\left(1 - {}_4p_{43}\right)$

$= {}_2p_{[40]}\, p_{43}\left(\frac{l_{43} - l_{47}}{l_{43}}\right)$

$= \underline{0.00671}$

(b) $_4p^{0.2}_{[40]+0.8} = p^{0.2}_{[40]}\ \underbrace{p_{[40]+1}\, p_{42}\, p_{43}}_{0.99631}\ p^{0.8}_{44}$

$= p^{0.2}_{36} \times 0.99631 \times p^{0.8}_{44}$

$= 0.994906$

$\therefore q = 1 - \text{''} = \underline{0.00509}$

(c) $\mathring{e}^{0}_{[40]+1} = \int_0^\infty t\, p_{[40]+1}\, dt = \int_0^1 t\, p_{[40]+1}\, dt + \int_1^\infty t\, p_{[40]+1}\, dt$

$= \int_0^1 \left(p_{[40]+1}\right)^t dt + p_{[40]+1} \int_0^\infty t\, p_{42}\, dt$

$= \int_0^1 (0.99902)^t\, dt + 0.99902\, \mathring{e}_{42}$

$= \left[\frac{0.99902^t}{\ln[0.99902]}\right]_0^1 + 0.99902\, \mathring{e}_{42}$

$$= \frac{0.99902 - 1}{\ln 0.99902} + 0.99902 \times 38.325$$

$$= 39.286$$

Then $\mathring{e}_{[40]}$, $= \frac{0.99925 - 1}{\ln 0.99925} + 0.99925 \times 39.286$

similarly,

$$= \underline{\underline{40.257}}$$

Q4 $\frac{e - l_{53.5} - l_{56.3}}{l_{53.5}}$

$$l_{53.5} = \frac{1}{2}(l_{53} + l_{54}) = \frac{1}{2}(9922.795 + 9914.586)$$

$$= 9918.79$$

$$l_{56.3} = 0.3 \times l_{57} + 0.7 l_{56}$$

$$= 6.002957$$

Q5

t	$x+t$	q_{x+t}	${}_tp_{60}$	$(v^{t+1})({}_tp_{60})(q_{x+t})$
0	60	0.03	1	0.028037
1	61	0.036	0.97	0.03050
2	62	0.0432	0.93508	0.032975
3	63	0.05184	0.894685	0.035383
4	64		0.848304	0.126846

$$A^{(4)}_{60:\overline{4}|} = 0.126896 \times (1.07)^{\frac{3}{8}}$$

$$= \cancel{0.127774}\ 0.130157$$

$$A_{60:\overline{4}|}{}^{1} = 0.848304\ v^4$$

$$= 0.647167$$

$$A^{(4)}_{60:\overline{4}|} = 0.127974 + 0.647167$$

$$= \cancel{0.775141}\ 0.777324$$

Q6

(a) Value of benefits $= 50{,}000\,\bar{A}^{\frac{1}{}}_{[50]:\overline{15}|} + 150{,}000\,\bar{A}^{\frac{1}{}}_{[50]:\overline{10}|}$

$$\bar{A}^{\frac{1}{}}_{[50]:\overline{15}|} = (1.04)^{\frac{1}{2}}\left[A_{[50]:\overline{15}|} - \frac{D_{65}}{D_{[50]}}\right]$$

$$= (1.04)^{\frac{1}{2}}\left[0.56695 - \frac{689.23}{1365.77}\right]$$

$$= 0.063538$$

Similarly $\bar{A}_{50:\overline{10}|} = 0.034325$

Value of benefits $= £8{,}325.60$

Value of premiums of £1 pam $= 12\,\ddot{a}^{(12)}_{[50]:\overline{15}|}$

$$\ddot{a}^{(12)}_{[50]:\overline{15}|} = \ddot{a}_{[50]:\overline{15}|} - \frac{11}{24}\left(1 - \frac{D_{65}}{D_{[50]}}\right)$$

$$= 11.302$$

$\therefore 12 \times 11.302\,P = £8325.60$

$P = £62.89$ per month

(b) $50{,}000\,\bar{A}^{0.04518}_{\frac{1}{}\,60:\overline{5}|} - 62.89 \times 12 \times \ddot{a}^{4.449\,(12)}_{60:\overline{5}|}$ [Ultimate]

$= -£1{,}099$

(c) Negative reserve; lapse $\Rightarrow$ loss to company

1) Shorten premium term
2) Make benefits more level.

(d) Actual death strain $= 9\,(200{,}000 - (-1099))$

$= £1,809,891$

Expected deaths $= 2000\, v_{57}^{.007140}$

$= 14.28$

EDS $= 14.28 \times (200{,}000 - (-1099))$

$= £2,871,693$

Profit $=$ difference

$= £1,061,802$

Q7

9

(a) T_x = time from age x to death, as a real number

K_x = Integer part of T_x

(b) (i) v^{T_x}

(ii) $a_{\overline{K_{20}}|}$

(iii) $v^{1+K_{[30]}}$ if $K_{[30]} \leq 9$

v^{10} if $K_{[30]} \geq 10$

(iv) $\ddot{a}_{\overline{K_x+1}|}$ if $K_x \geq 6$

0 if $K_x \leq 5$